Sagalpreet Singh

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Education

University of Oxford — Oxford, United Kingdom
DPhil in Computer Science, OATML Group; **Supervisor:** Prof. Yarin Gal — *2029 (tentative)*
Funded by LASR

Indian Institute of Technology Ropar — Rupnagar, India
Bachelor of Technology in Computer Science and Engineering; **CGPA:** 9.31/10 — *May 2023*
Concentration in Artificial Intelligence; **CGPA:** 10/10

Publications

C=Conference

[C.3] Dense and Diverse Goal Coverage in Reinforcement Learning
Sagalpreet Singh, Rishi Saket, Aravindan Raghuveer **[RLC '26 (Oral)]** [Paper]

[C.2] Learning from Label Proportions and Covariate-shifted Instances
Sagalpreet Singh, Navodita Sharma, Shreyas Havaldar, Rishi Saket, Aravindan Raghuveer **[UAI '25]** [Paper]

[C.1] On Subset Selection of Multiple Humans to Improve Human-AI Team Accuracy
Sagalpreet Singh, Shweta Jain, Shashi Shekhar Jha **[AAMAS '23 (Oral)]** [Paper]

Experience

Google DeepMind | *Pre-Doctoral Researcher* — *Aug 2024 – Jul 2026*

- Ran end-to-end **RL training** on **Gemini** models across **multi-step coding** and **long-horizon** environments; designed production-scale data mixtures; evaluated impact through systematic **checkpoint evaluation** and **auto-analysis** to attribute gains across algorithm, data, and infrastructure
- Monitored **training curves** and **trajectories** to detect **reward hacking**; wrote **agentic workflows** for trajectory analysis to identify capability gaps; designed **evaluation setups** that distinguish genuine gains from artifacts of task or reward design
- Built **long-horizon RL** training data for **filesystem exploration and navigation** across **multiple Gemini scales**; achieved $>$**6 pp** improvement on downstream **agentic evals** with headroom confirmed for larger models; in collaboration with **Code Strike** and **Recursive Self-Improvement** teams
- Engineered scalable, **fully-verifiable curriculum generation** engine for **Gemini Thinking** via RL; contributed **38% weighted data** in the Gemini thinking data revision; $>$**10 pp** improvement on Math and NL Reasoning; in collaboration with **Thinking Strike** team
- Designed RL algorithm with **convergence guarantee** and **sample complexity** for preventing mode collapse in multi-goal environments [C.3]; demo at **NeurIPS '25** (Google DeepMind booth); **patent** filed
- **Multi-million dollar** impact on Google Ads and **SOTA** on public datasets via novel **domain adaptation** objective for weakly supervised target domains [C.2]; derived generalization bounds and sample complexity

Oracle AI | *Member of Technical Staff* — *Jun 2023 – Aug 2024*

- Built Oracle's **transformer-based** in-house **Text-to-Speech model** with **few-shot voice cloning**; trained on **distributed Slurm cluster**; deployed via **Triton inference server**; achieved **SOTA WER/CER** against baselines

Honors & Awards

*=Declined

- **Google Research Travel Grant** (USD 2000) & **Microsoft Research Travel Grant*** (INR 120k)
- **NTSE Scholarship** by Government of India (top 1000 students nationally)
- **Applied Scientist Internship*** — Amazon (via ML Summer School)
- **Institute Merit Scholarship** (top 7% of batch) | **Best Poster** & **Best B.Tech. Project** — IIT Ropar

Academic Service

Reviewer: NeurIPS '26, ICML '26, UAI '26, AISTATS '26, ACL-ARR '25